

综合利用

利用七水硫酸亚铁生产一水硫酸亚铁及聚合硫酸铁

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摘要: 钛白粉生产中副产大量的七水硫酸亚铁, 采用合理工艺可以同时生产饲料级一水硫酸亚铁和水处理剂聚合硫酸铁, 并实现废水零排放。

关键词: 七水硫酸亚铁; 一水硫酸亚铁; 聚合硫酸铁

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钛白粉生产过程中形成大量副产物七水硫酸亚铁, 近年来随着我国饲料工业的发展, 大部分都用于生产饲料用硫酸亚铁。

我们通过多年的研究和生产实践, 探索出了一条综合利用七水硫酸亚铁生产饲料级一水硫酸亚铁和水处理剂聚合硫酸铁的最优化工艺。

1 原料和试剂

七水硫酸亚铁: 取自攀钢钛业公司钛白粉厂, Fe^{2+} 质量分数为 17%, As 为 0.000 2%, Pb 为 0.002%, 水不溶物 < 0.2%; 硫酸为工业品, 质量分数含量为 95%; 氯酸钾为工业品; 脱水剂自备。

2 试验方法

2.1 试验流程

图 1 为七水硫酸亚铁生产一水硫酸亚铁和聚合硫酸铁的流程简图。

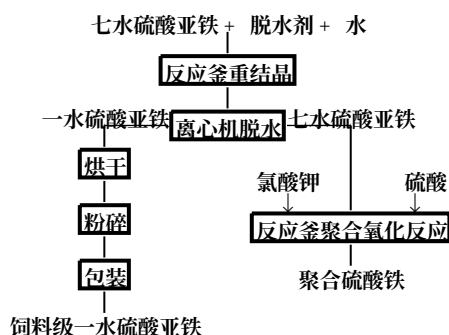


图 1 一水硫酸亚铁、聚合硫酸铁联合生产工艺流程简图

2.2 试验工艺

如流程图所示: 从制备一水硫酸亚铁重结晶工

艺开始, 先在反应釜内加入待反应晶体七水硫酸亚铁质量 10% 的水和脱水剂, 通入蒸汽加热, 开动搅拌, 然后投入七水硫酸亚铁, 投料完毕后继续搅拌升温反应, 到 60 °C 时开始熔化, 90 °C 时熔化完毕, 继续升温, 开始重结晶, 控制好温度和反应时间到重结晶反应完毕, 然后趁热放料离心脱水, 脱水后中间产品中仍含 10% 左右的游离水, 经干燥和粉碎后得到一水硫酸亚铁产品。

离心脱水的含酸和 Fe^{2+} 废水可返回小部分到重结晶工序; 其余大部分积聚到一定量后就可以投到反应釜作为生产聚合硫酸铁的原料, 再添加计量好的七水硫酸亚铁、硫酸、氯酸钾等, 并通氧反应, 可制得高效水处理剂聚合硫酸铁。

3 结果与讨论

3.1 重结晶温度试验

在重结晶反应过程中, 温度的控制非常重要, 虽然从机理上看, 在 64~90 °C 七水硫酸亚铁会失去 6 个结晶水, 但重结晶的温度却高于此温度, 也就是它虽然脱去了 6 个结晶水, 但是尚未形成 $\text{FeSO}_4 \cdot \text{H}_2\text{O}$ 结晶体, 很容易在温度降低时又变成七水硫酸亚铁晶体。只有重结晶完成后, 变成了一水硫酸亚铁晶体, 才能不会很快变成七水硫酸亚铁, 只要及时烘干一水硫酸亚铁中的游离水, 包装密封好, 就可以较长时间贮存。

温度对一水硫酸亚铁结晶体影响如图 2, 当达到熔化温度后, 随着温度的升高, 一水硫酸亚铁结晶体的含量不断增加, 当 110 °C 时就基本达到 80% 左右, 再升高温度, 结晶体增加不大, 即达到了饱和状

态。因此确定结晶最终反应温度为 110~120 °C。

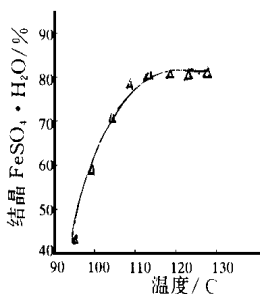


图2 温度对结晶 $\text{FeSO}_4 \cdot \text{H}_2\text{O}$ 含量的影响

3.2 脱水剂的影响

脱水剂对七水硫酸亚铁的重结晶脱水影响很大,不加脱水剂,离心所得中间产物水分含量高,紧密,烘干困难,且易于吸水恢复成七水硫酸亚铁,很难生产出合格的一水硫酸亚铁。根据试验,确定脱水剂用量为七水硫酸亚铁投料量的 1.4%~1.6%,用量过低脱水不充分,用量过高则加大了成本。

3.3 最终产品指标

按以上讨论确定一水硫酸亚铁重结晶工艺为温度 110~120 °C,脱水剂用量为 1.4%~1.6% 的七水硫酸亚铁投料量,反应时间控制在 0.5~1 h 内,可得到指标如表 1 的一水硫酸亚铁产品,完全符合饲料级标准要求。

表 1 一水硫酸亚铁各项指标

指标名称	Fe^{2+}	重金属(以 Pb 计)	砷	外观
含量/%	≥ 30	≤ 0.001	≤ 0.0005	灰白粉末

3.4 聚合硫酸铁

先分析废水中 Fe^{2+} 和 H_2SO_4 的含量,再按 1 t 聚合硫酸铁消耗 H_2SO_4 70 kg,七水硫酸亚铁消耗 580 kg 的总量要求,补加 H_2SO_4 和七水硫酸亚铁;反应时缓缓加入氧化剂氯酸钾 38 kg/t,并通入

空气氧化,经水解、聚合反应即得聚合硫酸铁,最终指标如表 2。聚合硫酸铁对冶金、火电、纺织印染、肉食加工、炼油等工业废水及城市综合废水、生活废水有良好的净化效果。

表 2 聚合硫酸铁各项指标

指标名称	Fe^{3+} $\text{g} \cdot \text{L}^{-1}$	Fe^{2+} $\text{g} \cdot \text{L}^{-1}$	粘度(20°C) $\text{MPa} \cdot \text{s}$	相对密度	碱化度 %
含量	≥ 160	≤ 0.1	8~15	1.45~1.5	10~15

4 经济分析

根据我们的生产经验,一水硫酸亚铁的成本构成为以下几项:七水硫酸亚铁和脱水剂 420 元/t;水、电、汽消耗 110 元/t;人员工资 100 元/t;包装 30 元/t;折旧 50 元/t;合计综合成本为 710 元/t,目前市场售价为 1 100 元/t,一水硫酸亚铁毛利 390 元/t。若按年产 2 000 t 估算,每年可创毛利 78 万元。

聚合硫酸铁因系利用废水生产,其成本较低,成本构成:氧化剂 190 元/t, H_2SO_4 40 元/t,七水硫酸亚铁 60 元/t,合计 290 元/t,目前市场售价为 650 元/t,毛利 360 元/t。若按年产 2 000 t 估算,每年创毛利 72 万元。与 2 000 t 一水硫酸亚铁所创造的效益基本相当;同时解决了废水的出路问题,变废为宝,社会效益也非常明显。

5 结论

选取图 1 的工艺,可以综合利用钛白粉副产物七水硫酸亚铁生产出合格的饲料级一水硫酸亚铁和优良的水处理剂聚合硫酸铁。如果该工艺由钛白生产厂家采用,还可综合利用钛白生产中产生的废酸(20% H_2SO_4),配合七水硫酸亚铁生产聚合硫酸铁;对于我国钛白生产实现无废排放具有重要意义。

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波强度增加至 600 W 后,产率反而下降,可能是高强度微波晶化时,产生少量缺陷、包藏所致,微波功率取 600 W 为宜。

3 结论

用微波诱导,由磷酸二氢钠和氯化铝合成磷酸铝,与经典的水热法合成磷酸铝相比,反应时间从 7 日缩短至 40 min,节省能量消耗,降低了生产成本。在本实验条件下,磷酸铝的最佳合成条件:磷酸二氢

钠与氯化铝的配比为 4:1,微波晶化时间为 40 min,微波功率为 600 W,产率可达 95.8%。

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high purity; phosphorus

SEPARATING KBr FROM THE AQUEOUS SOLUTION BY SOLVENTING OUT METHOD. WANG Jiancheng *et al* (Dept of Light Chemical Engineering, Shangdong Institute of Light Industry, Jinan 250100): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 25~26

Three organic solvents, methanol, ethanol and acetone are selected to separate KBr from the aqueous solution. It shows that KBr in aqueous solution can be separated effectively by methanol, ethanol and acetone, and the separating effect of methanol is better than that of other two. The experimental result can be described quite accurately by the equation $f = K \ln(V/\sqrt{c})$, for $f < 0.9$. The above equation can act as a basis of engineering design.

Key words: crystallization; separation; solventing out; potassium bromide

REDUCING ROAST OF STRONTIANITE WITH CARBON AT HIGH TEMPERATURE. ZHANG jin *et al* (Dept of Chemistry, Chongqing Teacher's College, Yongchuan 402168): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 27~28

The theoretical basis and experimental result to make strontium oxide from strontianite by carbon reduction of high temperature are introduced. A new way to solve the problem of sintering in vertical kiln is found.

Key words: strontianite; strontium oxide; strontium carbonate; carbon reduction; roast

THE PROCESS FOR PRODUCING MONOHYDRATE FERROUS SULFATE AND POLYFERRIC SULFATE FROM FERROUS SULFATE. GE yingyong *et al* (Changsha Research Institute of Mining and Metallurgy, Changsha 410012): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 29~30

A large amount of by-product ferrous sulfate is produced in the production of TiO_2 pigment, the technologies of producing feed-grade monohydrate ferrous sulfate and water treatment agent-polyferric sulfate from the above by-product simultaneously are presented.

Key words: ferrous sulfate; monohydrate ferrous sulfate; polyferric sulfate

STUDY ON THE PREPARATION OF K_2SO_4 FROM THE BY-PRODUCT FeSO_4 IN THE PRODUCTION OF TiO_2 . LIU Xiaohong *et al* (Dept of Chem. Eng., Nanchang Univ., Nanchang 330029): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 31~32

The K_2SO_4 is prepared by two-step double decomposition method, the by-product FeSO_4 from TiO_2 production and KCl are as raw materials. The effect of amounts of water adding, water evaporating and washing water on the process are studied.

Key words: ferrous sulfate; potassium sulfate

THE PREPARATION OF POLYALUMINUM CHLORIDE COAGULANT FROM KAOLINITE. SHEN shuifa *et al* (Institute of Functional Materials, Fuzhou Univ., Fuzhou 350002): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 33~35

A new method for preparation of polyaluminum chloride coagulant is studied. Kaolinite is used as aluminum source and other aluminum or basic materials are not required. The product is obviously superior to traditional coagulant in either purification of drinking water or sewage

treatment.

Key words: kaolinite; preparation; polyaluminum chloride

RECOVERY OF Na_2SO_3 MIXTURE FROM SULFUR-CONTAINING WASTEWATER AS THE SLURRY FOR PAPER-MAKING. ZOU changjun *et al* (Dept of Fine Chemicals, Jilin Institute of Chemical Technology, Jilin 132022): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 36~37

Na_2SO_3 mixture is prepared from sulfur-containing wastewater by concentration method and the mixture is used as slurring for paper-making. The recovery of Na_2S can reach 100%, the economic and environmental benefits are remarkable.

Key words: sulfur-containing wastewater; concentration method; slurry

DATA DISCUSSION FOR THREE-PHASE CIRCULAR ELECTRIC FURNACE OF PRODUCING PHOSPHORUS. CAI peijian (Yunnan Qujing Research and Design Institute of Chemical Industry, Qujing 655000): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 38~40

Based on the specific resistivity in the electric furnace of producing phosphorus, the basic electric and geometric data are discussed.

Key words: electric furnace of producing phosphorus; data; formula

THE INTERACTION BETWEEN THE QUALITY OF POWER-SUPPLYING AND THE EQUIPMENT RUNNING. DIN G liang *et al* (Ji'nan Yuxing Chemical General plant, Ji'nan 250033): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 41~43

In the industrial production, the safe, reliable and normal running of the equipments are very important for the realization of various process index. Meanwhile the good quality of power-supplying not only ensures the normal running of the equipment, but also has an effect on accuracy and reliability of lighting, controlling, protecting and detecting of the equipment. On the contrary, the quality of the equipment running can feed back to the power-supplying net and cause the effect on the other instruments.

Key words: sinusoidal a.c.; voltage variation; higher harmonic wave

THE DETERMINATION OF CHLORINE DIOXIDE BY SPECTROPHOTOMETRIC METHOD. ZHANG hongyun *et al* (School of Chemistry and Chemical Engineering, Zhengzhou University, Zhengzhou 450052): Chin. J. Inorganic Chemicals Ind 31(5), 1999, pp. 44~45

Using chlorophenol red as a spectrophotometric reagent for the determination of the chlorine dioxide is described. The max. absorption wave length is $\lambda = 574 \text{ nm}$, at $\text{pH} = 9 \sim 11$, the influence of acidity is small. In experimental procedure, first, the ammonia-ammonium chloride buffer solution is added. Then free chlorophenol red is added. When free chlorine concentration up to 22.86×10^{-6} , the determination of chlorine dioxide has no interference. The chlorophenol red spectrophotometric method combined with the oxidation reduction titration of ferrous ammonium sulfate, the chlorine dioxide and free chlorine can be simultaneously determined.

Key words: chlorophenol red; spectrophotometric method; chlorine dioxide; free chlorine; determination